Research Much interest has been aroused by eucommia bark's ability to reduce high blood pressure, which it is thought to do by increasing nitrous oxide levels within the arteries. In a clinical trial in China involving 119 people, 46% of those treated with the herb showed a significant blood pressure reduction. However, eucommia bark appears to have little effect in cases of severe hypertension. Recent studies indicate that eucommia bark is an antioxidant and may help to prevent the onset of type 2 diabetes. A small clinical trial in Japan published in 1996 concluded that an infusion of eucommia bark reduced the body's exposure to mutagen-forming compounds naturally present within the diet.

Euonymus atropurpureus (Celastraceae)

Wahoo Bark

Description Deciduous tree growing to 26 ft (8 m). Has smooth branches, serrated elliptical leaves, clusters of purple flowers, and 4-lobed scarlet fruit.
Habitat & Cultivation Native to eastern North America, wahoo bark thrives in damp woods and close to water. The bark is gathered in autumn.
Parts Used Stem bark, root bark.
Constituents Wahoo bark contains cardenolides (cardiac glycosides) similar to digitoxin, asparagine, sterols, and tannins.
History & Folklore The Sioux, Cree, and other Native American peoples used wahoo bark in various ways, for example as an eye lotion, as a poultice for facial sores, and for gynecological conditions. Native Americans introduced the plant to early European settlers, and it became very popular in North America as well as in Britain in the 19th century.
Medicinal Actions & Uses Wahoo bark is considered a gallbladder remedy with laxative and diuretic properties. It is prescribed for biliousness and liver problems, as well as for skin conditions such as eczema (which may result from poor liver and gallbladder function) and for constipation. In the past, it was often used in combination with herbs such as gentian (*Gentiana lutea*, p. 99) as a fever remedy, especially if the liver was under stress. Following the discovery that it contains cardiac glycosides, wahoo bark has been given for heart conditions.
Cautions Wahoo bark is toxic. Use only under professional supervision. Do not take during pregnancy or while breastfeeding.

Eupatorium cannabinum (Asteraceae)

Hemp Agrimony

Description Perennial growing to a height of 5 ft (1.5 m). Has a red stem, downy leaves, and dense bunches of pink to mauve florets.
Habitat & Cultivation Native to Europe, hemp agrimony is now also found in western Asia and North Africa. It grows in damp woods, ditches, marshes, and in open areas, and is gathered when in flower in summer.
Parts Used Aerial parts, root.
Constituents Hemp agrimony contains a volatile oil (with alpha-terpinene, p-cymene, thymol and an azulene), sesquiterpene lactones (especially eupatoriopicrin), flavonoids, pyrrolizidine alkaloids, and polysaccharides. P-cymene is antiviral, while eupatoriopicrin has anti-cancer properties and inhibits cellular growth. The polysaccharides stimulate the immune system. In isolation, the pyrrolizidine alkaloids are toxic to the liver.
History & Folklore Hemp agrimony was known to Avicenna (980–1037 CE) and other practitioners of Arabian medicine in the early Middle Ages.

Hemp agrimony *was formerly taken as a spring tonic in Holland.*

Medicinal Actions & Uses Hemp agrimony has been employed chiefly as a detoxifying herb for fever, colds, flu, and other acute viral conditions. The root is laxative and diuretic, and the whole herb is considered to be tonic. Recently, hemp agrimony has found use as an immunostimulant, helping to maintain resistance to acute viral and other infections.
Related Species See also boneset (*E. perfoliatum, following entry*) and gravel root (*E. purpureum, subsequent entry*).
Caution In view of hemp agrimony's pyrrolizidine alkaloid content, take only under professional supervision.

Eupatorium perfoliatum (Asteraceae)

Boneset

Description Erect perennial growing to 5 ft (1.5 m). Has tapering lance-shaped leaves and many white or purple florets.
Habitat & Cultivation Native to eastern North America, boneset is found in meadows and marshland. It is gathered when in flower in summer.
Parts Used Aerial parts.
Constituents Boneset contains sesquiterpene lactones (including eupafolin), polysaccharides, flavonoids, diterpenes, sterols, and volatile oil. The sesquiterpene lactones and polysaccharides are significant immunostimulants.
History & Folklore Native American people used boneset to make an infusion for treating colds, fever, and arthritic and rheumatic pain. European settlers learned of the plant's benefits, and by the 18th and 19th centuries it was regarded as a virtual cure-all. Boneset's common name derives from its ability to treat "break-bone fever." Commonly used to treat malaria, constituents within boneset are now known to have antiprotozoal activity.
Medicinal Actions & Uses A hot infusion of boneset will bring relief to symptoms of the common cold. The plant stimulates resistance to viral and bacterial infections, and reduces fever by encouraging sweating. Boneset also loosens phlegm and promotes its removal through coughing, and it has a tonic and laxative effect. It has been taken for rheumatic illness, skin conditions, and worms.
Related Species Wild horehound (*E. teucrifolium*) was used as a substitute for boneset. *E. occidentale* was used by the Zuni of the south-western U.S. to treat rheumatism. See also hemp agrimony (*E. cannabinum, preceding entry*) and gravel root (*E. purpureum, following entry*).
Self-help Uses Allergic rhinitis with mucus, p. 300; Colds, flu & fevers, p. 311; High fever, p. 311.